



FUNDAMENTALS OF HUMAN COMPUTER INTERACTION

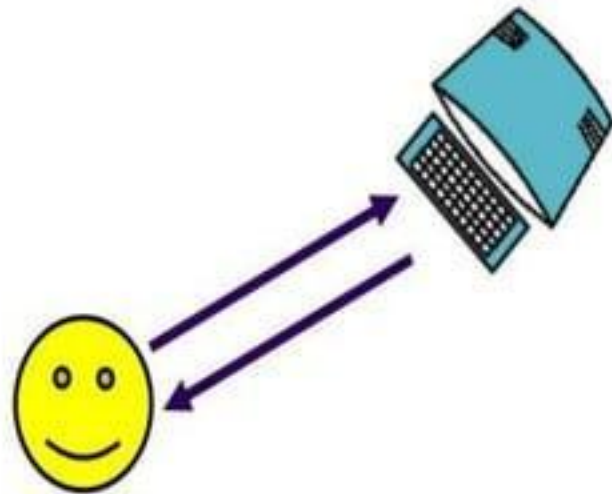
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What is HCI ?

- Point of communication between the user and the computer
- Studies the design and use of computer technology focused on the interface between the user and the system

Human Computer Interaction has Three Components

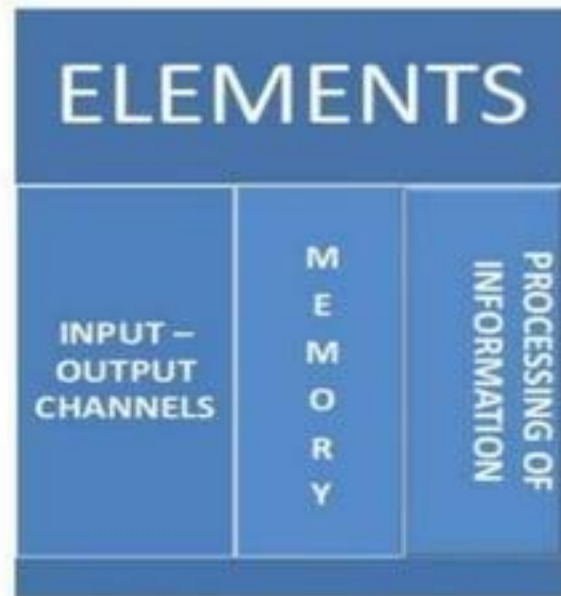
- ▶ Human
- ▶ Computer
- ▶ Interaction



The goal of HCI is to improve the interaction between users and computers by making computers more user-friendly and receptive to the user's needs.

Overview

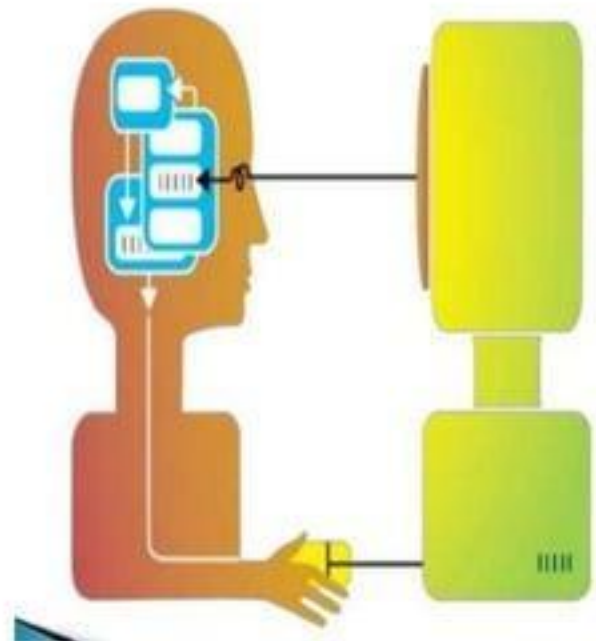
- Interaction with world
 - Occurs through information
- Interaction with computer
 - Input and output channels



- Then the information is stored in the memory
- Finally the information is processed and applied.
 - Reasoning
 - Problem solving
 - Skill acquisition
 - Error

Input-Output Channels

- Human interacts by sending and receiving information
- Human Input
 - Through Senses
- Human Output
 - Through Effectors





Input via Senses

- Vision
- Hearing
- Touch
- Taste
- Smell



Output via Effectors (Responders)

- Limbs
- Fingers
- Eyes
- Head
- Vocal system



Interaction with PC Using Input-Output Channels

- Using a GUI-based computer
- Information received by sight
- Beeps received by ear
- Feel keyboard and mouse with fingers



3 Major elements of Interaction

- Vision
- Hearing
- Touch

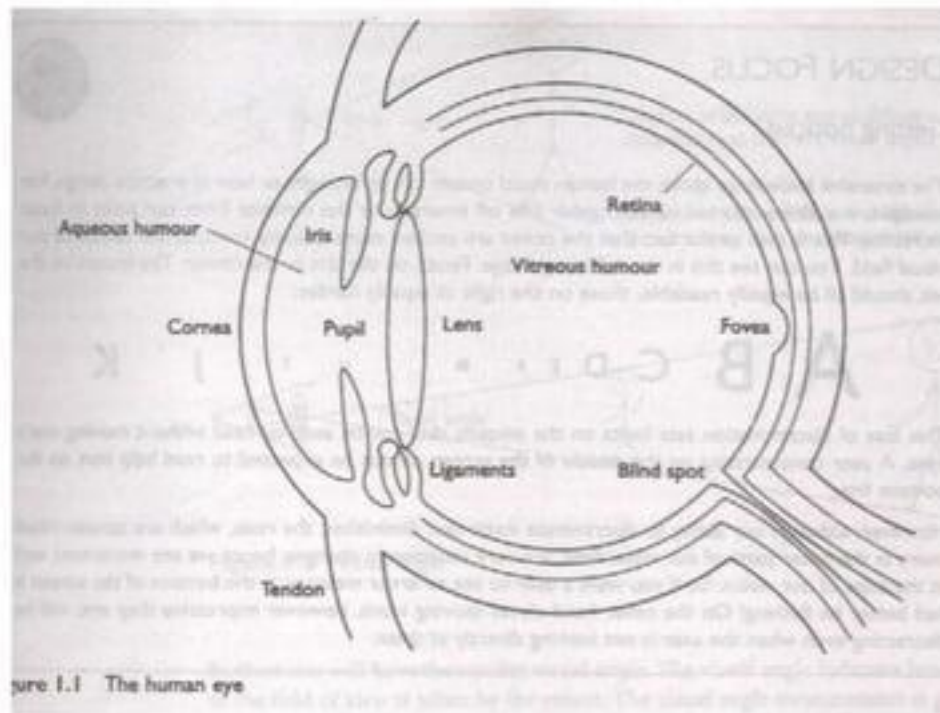
Vision

- Human Eye
- Visual Perception
- Reading

Vision

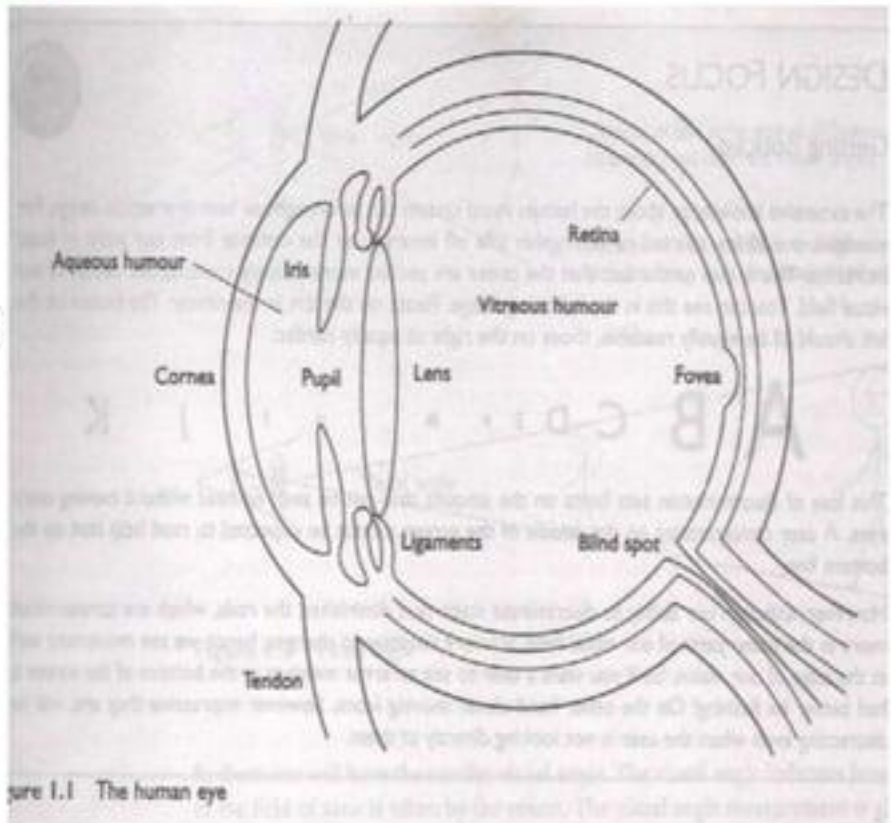
- Highly complex activity
- Physical and perceptual limitations
- 2 stages of visual perception:
 - Physical reception of the stimulus
 - Interpretation/processing of stimulus
- Processing allows construction of images from incomplete information

Human Eye



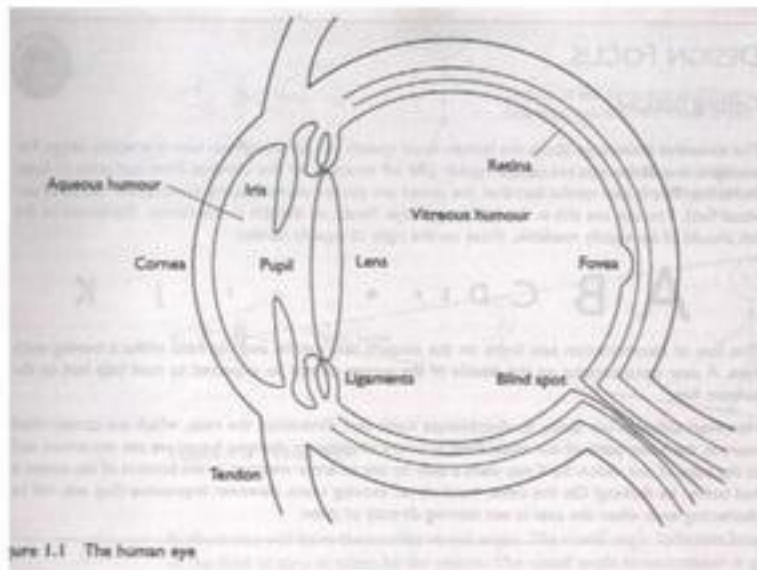
Human Eye Image Formation

- Cornea and lens
 - Focuses light into a sharp image on retina
 - An upside down image is formed on the retina.



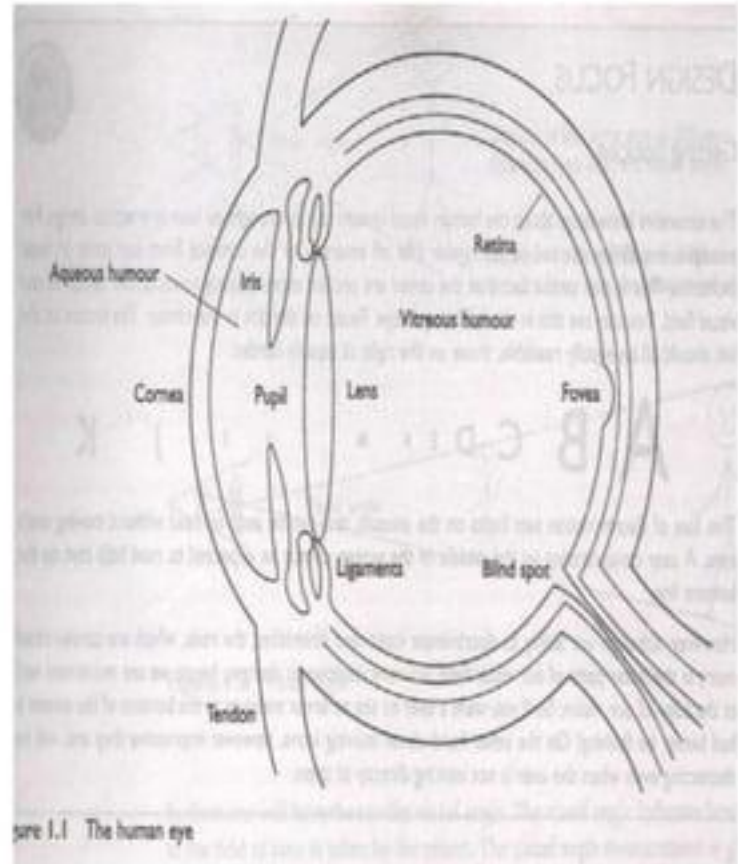
Photoreceptors

- Rods
- Cones



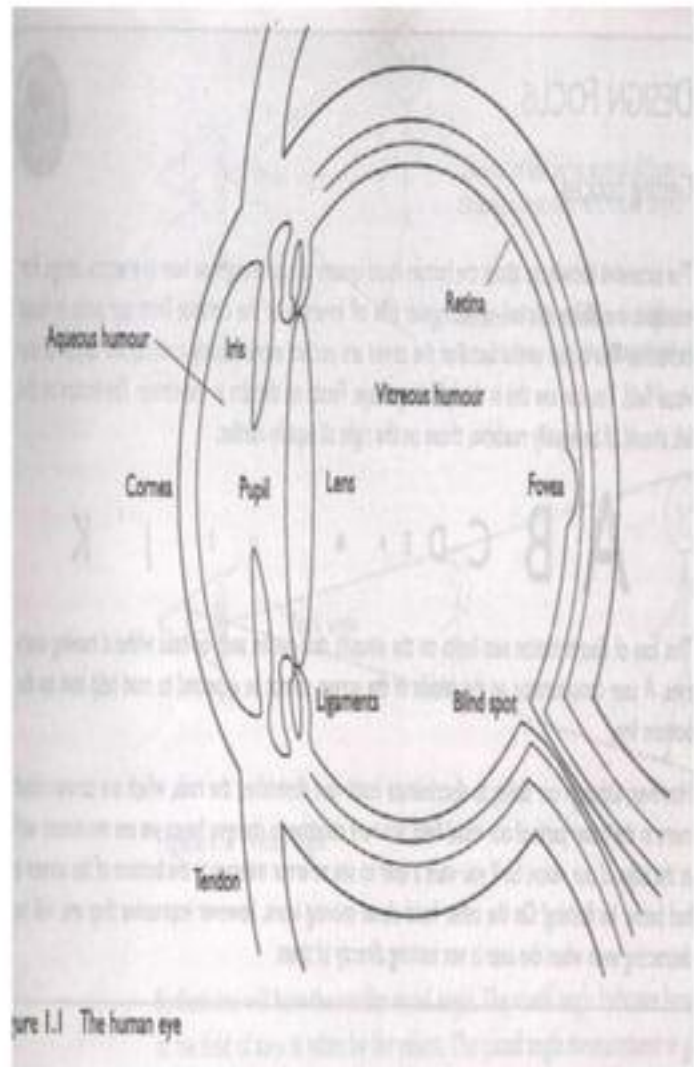
Rods

- Situated towards the edges of retina
 - Dominate peripheral vision
- Highly Sensitive to light
 - Allow us to see under low level of illumination
- Unable to resolve fine detail and are subject to light saturation
 - Cause of temporary blindness when moving from dark areas to very bright ones
- 120 million rods per eye



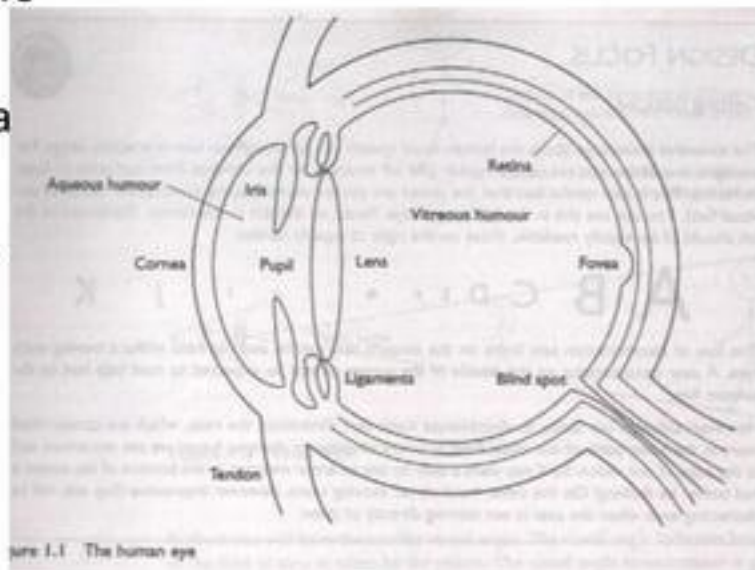
Cones

- Less sensitive to light
 - Can tolerate more light than Rods
- Basic function is color vision
- Situated in Fovea
 - Small area on retina where image is fixated
- Three types of cones
 - Each sensitive to a different wavelength



Blind Spot

- Area where optic nerve enters
- No rods or cones in this area
- Visual system compensates for lack of rods and cones



Nerve Cells

- A.k.a. Ganglion Cells
- Types
 - X-cells
 - concentrated in fovea
 - detection of patterns
 - Y-cells
 - widely distributed in retina
 - early detection of movement
 - can not detect change in patterns

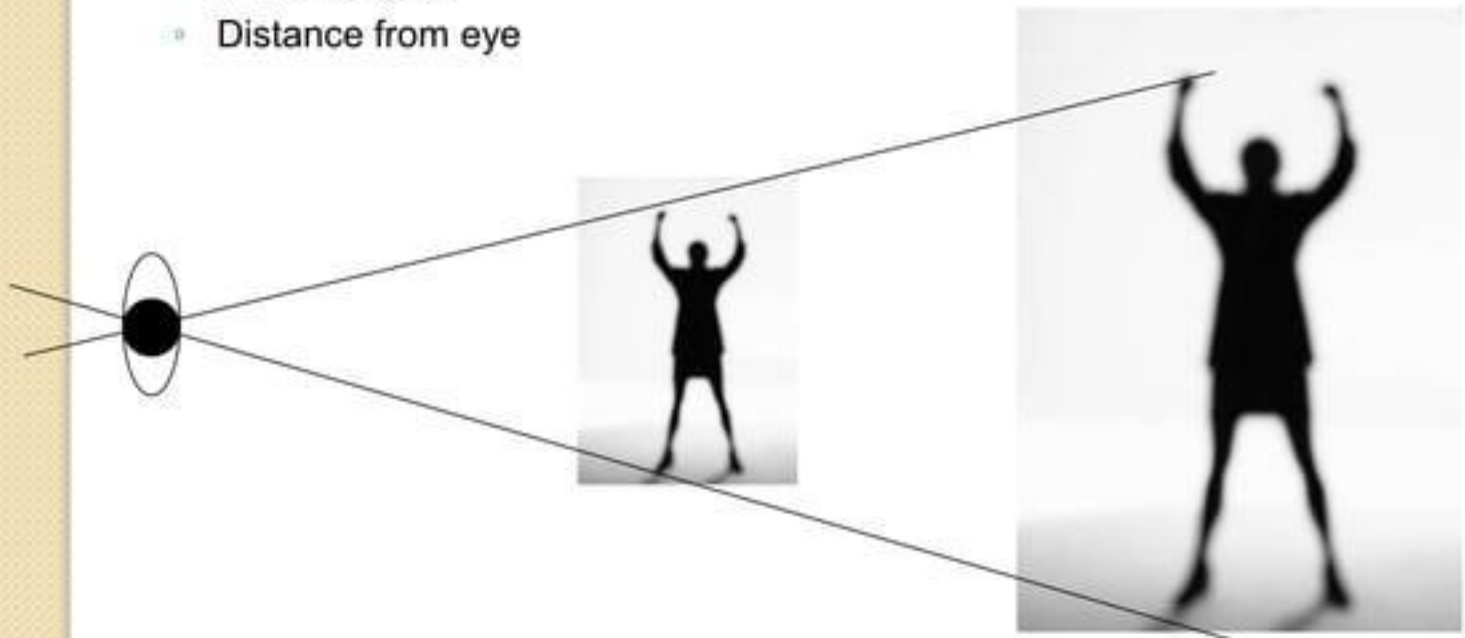


Notions of Size and Distance

- You are standing on a hill
- Rocks, sheep and small tree on summit
- Farmhouse on hillside
- Person walking on track
- Small market town in valley

Perceiving Size and Depth

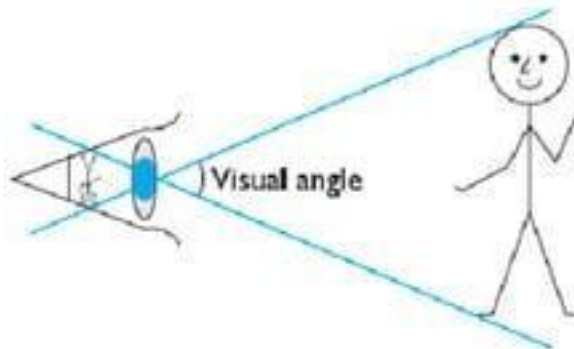
- Size specified by visual angle
- Affected by both
 - Size of object
 - Distance from eye



Perceiving Size and Depth

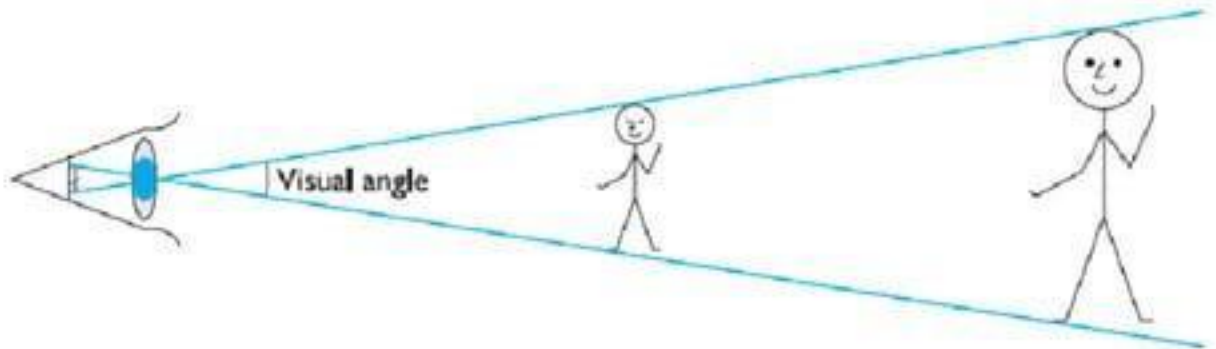
- Visual angle

- Indicates how much of the field of view is taken by the object
- Measured in degree or minutes of arc



Objects of the same size at different distances have different visual angles

Objects of different sizes and different distances may have the same visual angle



Visual Angle and Perception

- Visual Acuity

- Visual Acuity is the ability of a person to perceive fine detail

- Law of size constancy

- Perception of the object size remains constant even if its visual angle changes
- Perception depends on factors other than visual angle



Factors Affecting Visual Perception

- Perception of depth
 - Cues to determine relative positions of objects
- Size and height of object
 - Provides cue for distance
- Familiarity
 - Certain size helps to judge the distance accordingly

Example

• A B C D E F . H I J K

Perceiving Brightness

- Brightness

- Is subjective reaction to levels of light
- affected by luminance of object

- Luminance

- Depends on
 - Amount of light falling on object
 - Reflective properties of object
- Measured by photometer

- Contrast

- Function of the luminance of an object and the luminance of its background

Perceiving Brightness

- Measured by just noticeable difference caused by luminance
- Visual acuity increases with luminance

Perceiving Color

- 3 components
 - Hue
 - Determined by the spectral wavelength
 - Blue (short)
 - Green (medium)
 - Red (long)
 - 150 hues determined by eye
 - Intensity
 - Brightness of color
 - Saturation
 - Amount of whiteness in the color
- 7 million colors can be perceived

Perceiving Color

- Color perception best in fovea, worst at periphery where rods are predominate
- 3-4% cones in fovea sensitive to blue light



Capabilities and Limitations of Visual Processing

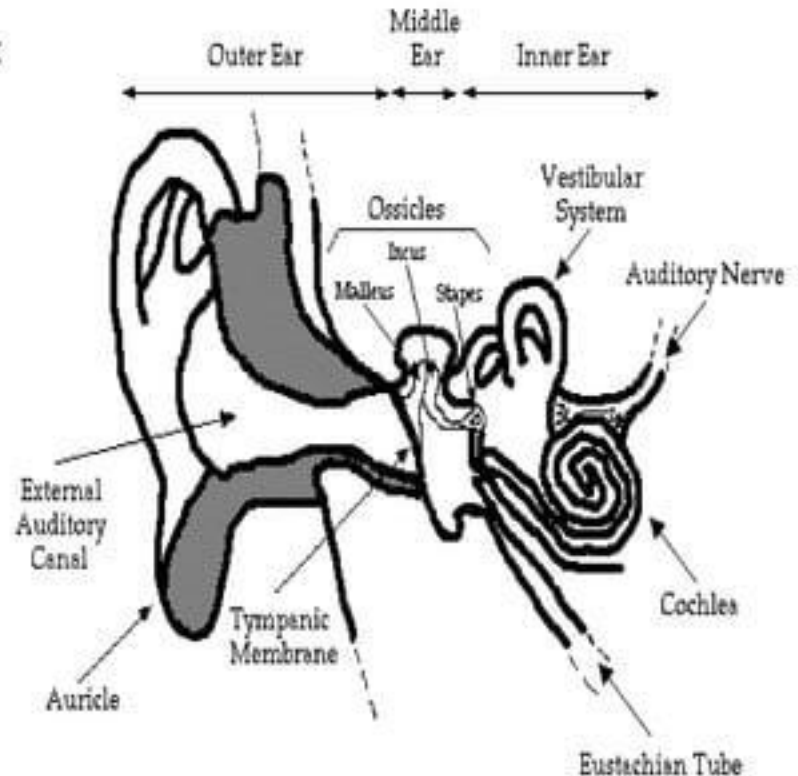
- Visual processing allows transformation and interpretation of a complete image
- Visual processing compensates image movement
 - Image moves on retina, but we see it stable
- Color and brightness perceived constant (hue, intensity, saturation)

Hearing

- Provides information about environment :

Human Ear and its components,

- **outer ear**
protects inner and amplifies sound
- **middle ear**
transmits sound waves as vibrations to inner ear
- **inner ear**
chemical transmitters are released and cause impulses in auditory nerve

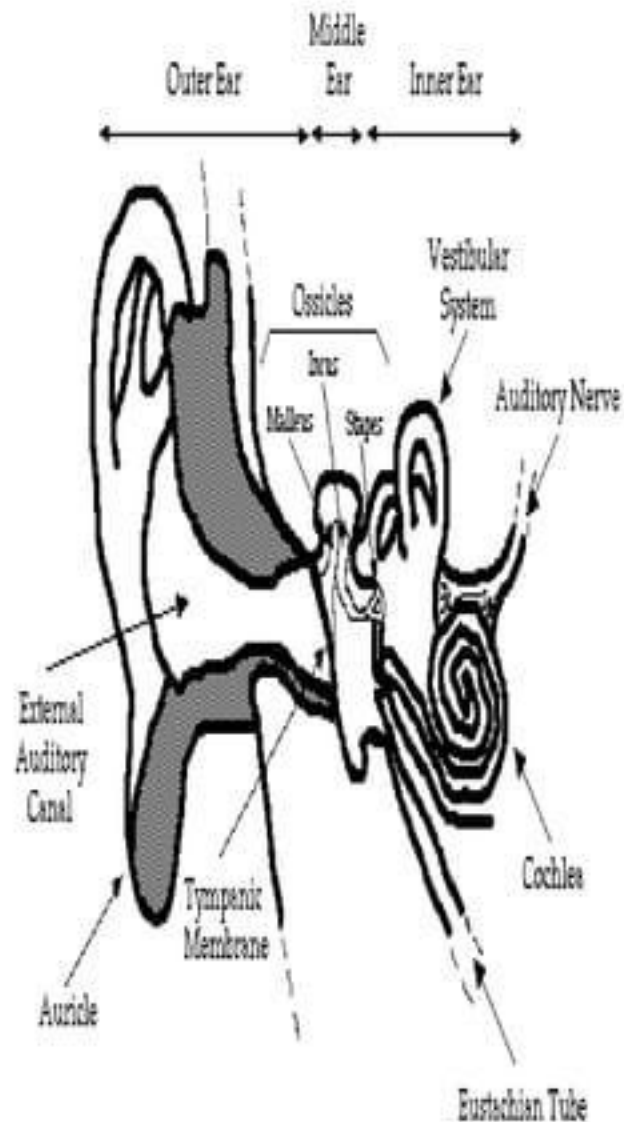


outer ear

- two parts:
 - *Pinna* - attached to the sides of the head
 - *auditory canal* - passes the sound waves to the middle ear.
- auditory canal contains wax which prevents dust, dirt and over-inquisitive insects reaching the middle ear

Middle Ear

- Passes the Sound waves along the auditory canal and vibrate the ear drum which in turn vibrates the ossicles, which transmit the vibrations to the cochlea, and so into the inner ear.



Inner Ear

- The waves are passed into the liquid-filled cochlea
- The delicate hair cells or *cilia bends because of the vibrations in the cochlean liquid* and release a chemical transmitter which causes impulses in the auditory nerve.

Processing sound

- Characteristics of Sound:
 - **Pitch** - frequency of the sound
 - **Loudness** - amplitude of the sound
 - **Timbre** - type of the sound

- 
- Humans can hear frequencies from 20Hz to 15kHz
 - Different frequencies trigger activity in neurons in different parts of the auditory system, and cause different rates of firing of nerve impulses
 - Auditory system filters sounds
 - Allows to ignore background noise and concentrate on important information

Touch

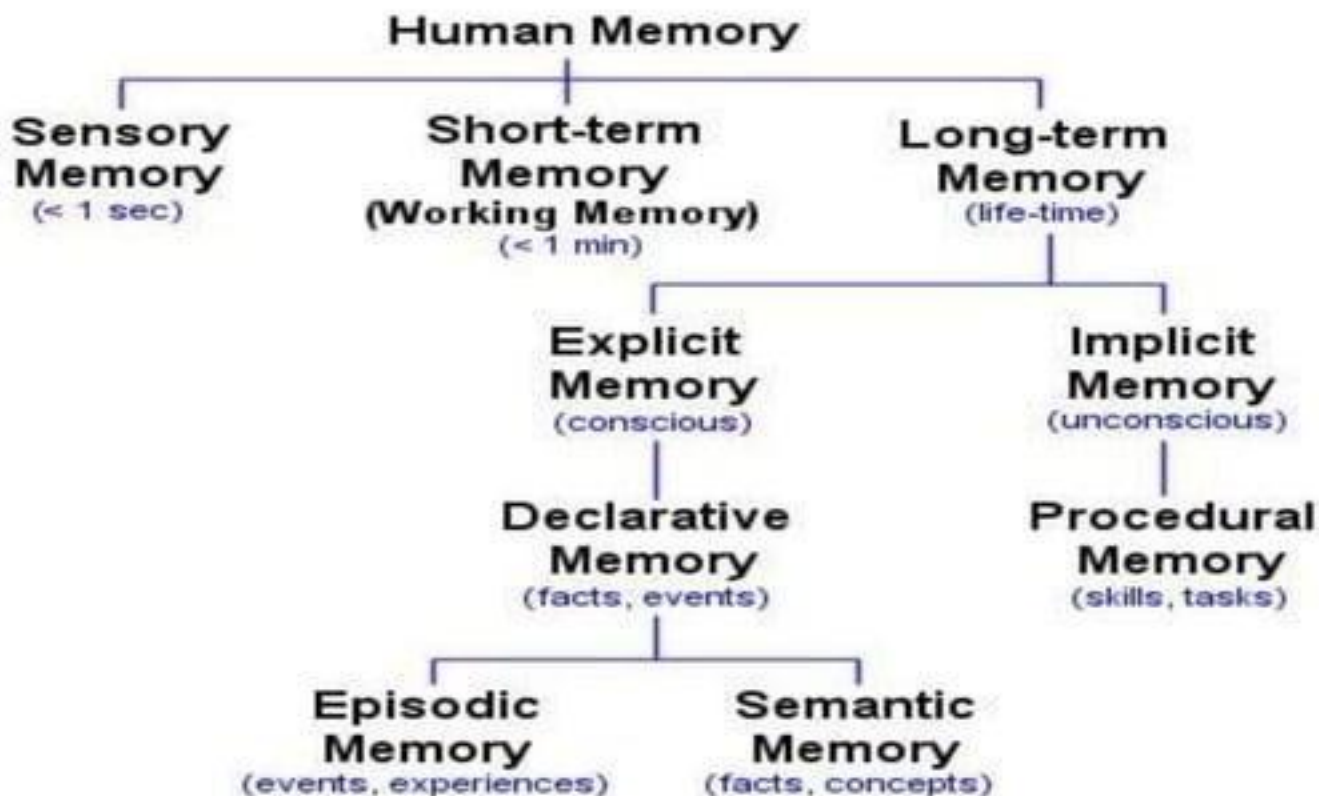
- last of the senses is touch or haptic perception.
- Provides vital information about our environment.
- It tells us when we touch something hot or cold, and can therefore act as a warning.
- Stimulus received via receptors in the skin

Types of sensory receptor

- Thermoreceptors
 - *Respond to heat and cold*
- Nociceptors
 - *Respond to pressure, heat and pain*
- Mechanoreceptors
 - *Respond to pressure*
 - *2 types:*
 - *Rapidly adapting mechanoreceptors - respond to immediate pressure*
 - *Slowly adapting mechanoreceptors - respond to continuously applied pressure*

- 
- Kinethesis - awareness of position of body & limb
 - affects comfort and performance.
 - Three types:
 - **rapidly adapting** - respond when a limb is moved in a particular direction;
 - **slowly adapting** - respond to both movement and static position;
 - **positional receptors** - respond only when a limb is in a static position.

Human Memory



Sensory Memory

- act as buffers for stimuli received through the senses
 - *iconic memory for visual stimuli,*
 - *echoic memory for aural stimuli*
 - *haptic memory for touch.*
- These memories are constantly overwritten by new information coming in on these channels.
- Examples – iconic memory
 - Moving a finger in front of the eye
 - “sparkler”
 - *Information remains in iconic memory very briefly, in the order of 0.5 seconds.*

- 
- Examples – *echoic memory*
 - direction from which a sound originates
 - Information is passed from sensory memory into short-term memory by attention, thereby filtering the stimuli to only those which are of interest at a given time.
 - Information received by sensory memories is quickly passed into a more permanent memory store, or overwritten and lost.

Short-term memory

- working memory
- acts as a 'scratch-pad' for temporary recall of information.
- used to store information which is only required fleetingly
- Example:
 - calculate the multiplication 35×6 in your head
 - 5×6 and followed by 30×6

- 
- Short-term memory can be accessed
 - rapid access $\sim 70\text{ms}$
 - rapid decay $\sim 200\text{ms}$
 - Two basic methods for measuring memory capacity.
 - determining the length of a sequence which can be remembered in order. **limited capacity 7 ± 2 chunks**
 - allows items to be freely recalled in any order.

Examples

212348278493202

0121 414 2626

HEC ATR ANU PTH ETR EET

chunking information can increase the
short-term memory capacity

Long Term Memory

- store factual information, experiential knowledge, procedural rules of behavior – Stores everything we know.
- Characteristics:
 1. It has huge capacity
 2. It has a relatively slow access time of approximately a tenth of seconds.
 3. Forgetting occurs more slow in long- term memory

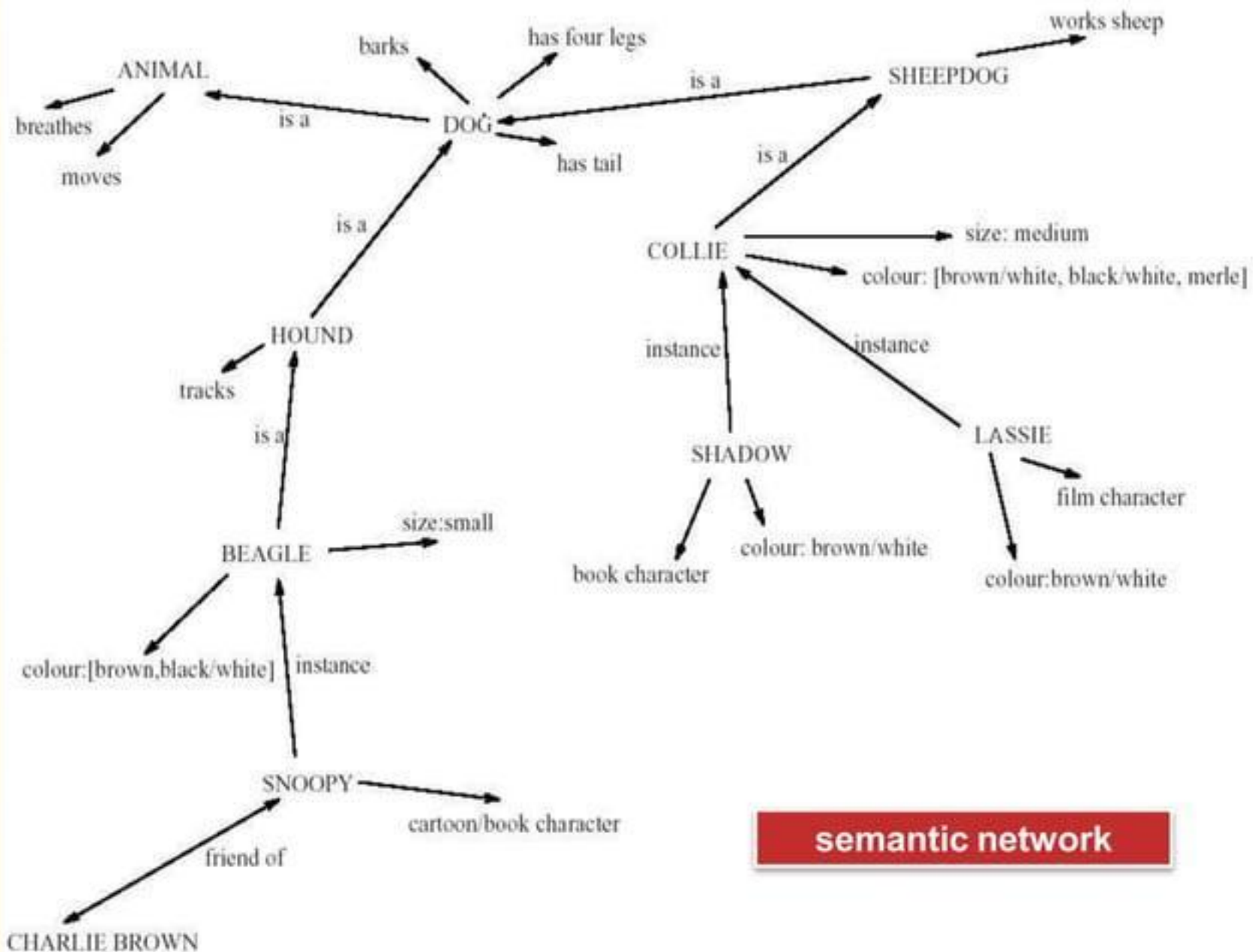
Long-term memory structure

- 2 types
 - *episodic memory*
 - represents memory of events and experience in a serial form.
 - Can recall an actual events took place at a given point of our lives.
 - *semantic memory*
 - structured record of facts, concepts and skills that we have acquired.
- semantic LTM derived from episodic LTM

LTM MODELS:

Semantic Network

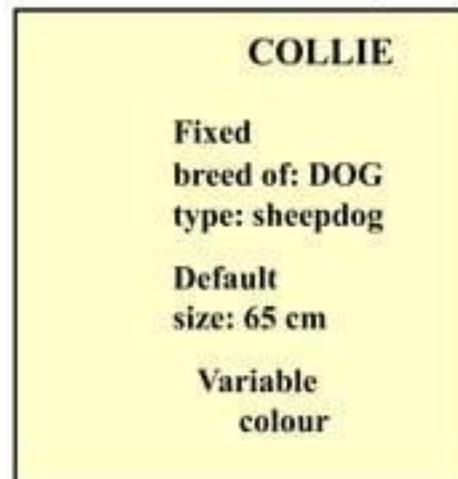
- Semantic memory is structured as a network.
- Items are associated to each other in classes, and may inherit attributes from parent classes.
- Example:
knowledge about dogs may be stored in a network as shown



LTM MODELS:

Frames

- Information organized in data structures
- Type-subtype relationships



- 
- Frame slots may contain default, fixed or variable information.
 - A frame is instantiated when the slots are filled with appropriate values.
 - Frames and scripts can be linked together in networks to represent hierarchical structured knowledge.

LTM MODELS: Scripts

- Scripts attempt to model the representation of stereotypical knowledge about situations.
- Ea: knowledge of the activities of doo

Script for a visit to the vet

Entry conditions:	<i>dog ill vet open owner has money</i>	Roles:	<i>vet examines diagnoses treats owner brings dog in pays takes dog out</i>
Result:	<i>dog better owner poorer vet richer</i>	Scenes:	<i>arriving at reception waiting in room examination paying</i>
Props:	<i>examination table medicine instruments</i>	Tracks:	<i>dog needs medicine dog needs operation</i>



A script comprises a number of elements, which, like slots, can be filled with appropriate information:

- **Entry conditions** Conditions that must be satisfied for the script to be activated.
- **Result** Conditions that will be true after the script is terminated.
- **Props** Objects involved in the events described in the script.
- **Roles** Actions performed by particular participants.
- **Scenes** The sequences of events that occur.
- **Tracks** A variation on the general pattern representing an alternative scenario.

LTM MODELS: Production rules

- Representation of procedural knowledge.
- Condition/action rules
 - if condition is matched
 - then use rule to determine action

IF dog is wagging tail THEN pat dog

IF dog is growling THEN run away

Long-term memory processes

- 3 main activities
 - storage or remembering of information,
 - Forgetting
 - information retrieval

Storage of information

rehearsal :

- Information is moved from short-term memory to long-term memory.
 - by repeated exposure to a stimulus or the rehearsal of a piece of information transfers it into long-term memory.
- repetition is not enough to learn information well. If information is not meaningful it is more difficult to remember.
- structure, meaning and familiarity
 - information easier to remember

Forgetting

- 2 main theories of forgetting:
 - *Decay*
 - *Interference.*

Decay

- information is lost gradually but very slowly

Interference

- new information replaces old: **retroactive interference**
- Ex: remembering your new phone number
- old may interfere with new: **proactive inhibition**
- Ex: find your self going to your old house instead of new one.

retrieval

- recall

- information reproduced from memory can be assisted by cues, e.g. categories, imagery

- recognition

- information gives knowledge that it has been seen before
- less complex than recall since the information is provided as cue



Thinking

Reasoning

deduction, induction, abduction

Problem solving

Reasoning

- Is the process by which we use the knowledge to draw conclusions or infer something new about the interest.
- inferring new information from what is already known
- Kinds of Reasons:
 - Deductive
 - Inductive
 - Abductive

Deductive Reasoning

- Deductive reasoning derives the logically necessary conclusion from the given premises.

e.g

- If it is Friday then she will go to work
It is Friday
Therefore she will go to work.

e.g. People from Pampanga cooks well and delicious
She is from Pampanga
Therefore she cooks well and delicious

Deductive Reasoning

- Logical conclusion not necessarily true:

e.g.

If it is raining then the ground
is dry It is raining
Therefore the ground is dry

Deduction (cont.)

- When truth and logical validity clash ...
e.g. Some people are babies Some babies
cry

Inference - Some people cry

Correct?

- People bring world knowledge to bear

Inductive Reasoning

- Induction:
generalize from cases seen to infer
information about cases unseen

e.g. all elephants we have seen have trunks therefore we
infer that all elephants have trunks.

- Unreliable:
 - can only prove false not true
- Humans not good at using negative evidence
e.g. Wason's cards.

Wason's cards



If a card has a vowel on one side it has an even number on the other

Is this true?

How many cards do you need to turn over to find out?

.... and which cards?

In fact, to test the truth of the statement we need to check negative evidence

Abductive reasoning

Abduction reasons from a fact to the action or state that caused it.

e.g.

Sam drives fast when drunk.

If I see Sam driving fast, assume drunk.

- **Unreliable:**

- can lead to false explanations

- If an event always follows an action, the user will infer that the event is caused by the action unless evidence to the contrary is made available.

- If, in fact, the event and the action are



Problem solving

- Process of finding solution to unfamiliar task using knowledge.
- There are a number of different views of how people solve problems.
- Several theories.

Gestalt Theory

- Problem solving is a matter of reproducing known responses or trial and error.
- problem solving both **productive** and **reproductive**
- Reproductive problem solving draws on previous experience.
 - Hindrance to finding a solution
- Productive problem solving involves insight and restructuring of the problem

Maier's *pendulum problem*







Problem space theory

- Proposed by Newell and Simon
- problem space comprises of problem states
- problem solving involves generating states using legal operators
- The problem has an initial state and a goal state and people use the operators to move from initial to the goal.
- heuristics may be employed to select operators

Sample Heuristic

means-ends analysis

- the initial state is compared with the goal state and an operator is chosen to reduce the difference between the two.
- Eg: reorganizing your office and you want to move your desk from the north wall of the room to the window
- Operators : carry or push or drag them
- If desk is heavy then new subgoal: to make it light.
- Operators : removing drawers and so on

- 
- – operates within human information processing system
 - e.g. STM limits etc.
 - – largely applied to problem solving in well-defined areas
 - e.g. puzzles rather than knowledge intensive areas

Problem solving (cont.)

- **Analogy**

- **analogical mapping:**
 - **novel problems in new domain?**
 - **use knowledge of similar problem from similar domain**
- **analogical mapping difficult if domains are semantically different**

- **Skill acquisition**

- **skilled activity characterized by chunking**
 - **lot of information is chunked to optimize STM**
- **conceptual rather than superficial grouping of problems**
- **information is structured more effectively**

Errors and mental models

Types of error

- **slips**

- right intention, but failed to do it right
- causes: poor physical skill, inattention etc.
- change to aspect of skilled behaviour can cause slip

- **mistakes**

- wrong intention
 - cause: incorrect understanding
- humans create mental models to explain behaviour.
if wrong (different from actual system) errors can occur



Thank You